Fernando Fernández Andueza

Gameplay and AI Programmer

andueza013.github.io/PortfolioGameDev anduezadev@gmail.com linkedin.com/in/fernando-fernandez-andueza anduezadev.itch.io
Valencia, Spain. EU citizen, open to relocation and to remote work. Spanish (native), English (intermediate).

Profile

Game programmer focused on game AI, gameplay systems and procedural generation, working in C++ with Unreal Engine 5 and in C# with Unity. I have a Higher National Diploma (HND) in Computing: Video Games Programming from ESAT and spent two and a half years as a fullstack developer before moving into games.

Technical skills

Engines and AI: Unreal Engine 5 (UE5) with C++ and Blueprints, Unity (C#), StateTree, Behavior Trees, EQS (Environment Query System), AI Perception, Animation Montages

Programming languages: C++ (C++17, C++20, C++23, STL), C#, C, ARM assembly (32- and 64-bit), Lua

Graphics and systems: OpenGL 4.5 DSA, deferred shading, PBR, vertex and fragment shaders, Entity Component System (ECS), multithreading and job systems, memory management

Tools: Visual Studio debugger, Unreal Insights, RenderDoc, Git, Perforce, Plastic SCM

Outside games: TypeScript, JavaScript, HTML, CSS, Vue.js, SQL Server, Java, VB.NET

Selected projects

Beerserker: The Stolen Brew — Unreal Engine 5, C++

Final-year project at ESAT, made with Grumpy Games by a team of about 20. Free and available on Steam.

- Built the four enemy types (melee, ranged, explosive and a tank mini-boss) in StateTree, with custom tasks and conditions in C++ and Blueprints.
- Wrote an object pool for enemies, projectiles, particles and decals, packaged as a plugin so other projects can reuse it.
- The ranged enemy uses EQS to find somewhere to retreat to, based on distance to the player and line of sight.
- The tank favours the melee attacks it has used least, so it doesn't keep repeating the same one.
- Built the enemy spawners, set up and uploaded the Steam builds, and took part in designing the architecture of a custom action system based on GAS (Gameplay Ability System).

XEMA (custom C++ engine) — C++, OpenGL 4.5

- Built from scratch on my own. Includes an ECS, deferred PBR rendering, shadow mapping, instancing, a job system that loads meshes asynchronously, Lua scripting and PhysX.
- Ported to Nintendo Switch (tested only on an emulator). PS5 port in progress.

Experience

Gameplay Programmer, Oniric Tales — Dec 2024 – Jun 2025

Roadtrip: The Engine of Madness (Unity)

- Built the core architecture of the game.
- Built the AI with Unity's Behavior Graph, including a perception system written entirely by me.
- Built a component-based dialogue system and the car radio, which plays songs on several stations like a real car radio.
- Also worked on other gameplay mechanics, profiling and bug fixing.

Fullstack Developer, AHORA — Sep 2020 – Mar 2023

- Developed business management software with FlexyGo, the company's in-house low-code framework, using C#, VB.NET, SQL Server, JavaScript, jQuery, HTML and CSS.
- Worked on a production plant's back office, the forklift drivers' handheld app, quality-control tools and an offline mobile clock-in app with facial recognition. Built a ZPL label-printing app on my own.
- Also worked on apps for maintenance, HR, sales and real estate management. Handled client incidents, often on site, in a Scrum team.

Education and certifications

HND in Computing: Video Games Programming, ESAT (Escuela Superior de Arte y Tecnología), Valencia — 2023–2026

C++, algorithms and data structures, memory management, low-level programming in 32- and 64-bit ARM assembly, graphics programming, and Unreal Engine and Unity. Final-year project: Beerserker.

DAM (Multiplatform Application Development), IES El Grao, Valencia — 2018–2020

Java for desktop and Android, relational databases, HTML and CSS, OOP fundamentals and version control.

Certified Scrum Master and Product Owner, Scrum.org / International Agile Institutes